

MTH109

Examination - January Semester 2026

Calculus

Tuesday, 21 April 2026

01:00 pm - 03:00 pm

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This examination contains **FIVE (5)** questions and comprises **FOUR (4)** printed pages (including cover page).
2. You must answer **ALL** questions.
3. All answers must be written in the answer book. Marks will only be awarded if **FULL** working is shown.
4. This is an **Open-book** examination.

At the end of the examination.

Please ensure that you have written your examination number on each answer book used.

Failure to do so will mean that your work cannot be identified.

If you have used more than one answer book, please tie them together with the string provided.

**THE UNIVERSITY RESERVES THE RIGHT NOT TO MARK YOUR
SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.**

(Full marks: 100)

Question 1

Determine the following limits.

Question 1a

$$\lim_{x \rightarrow -2} \frac{3x^3 - x^2 + 7}{5 - 4x}$$

(5 marks)

Question 1b

$$\lim_{x \rightarrow 22} \frac{\sqrt{26} - \sqrt{x + 4}}{x - 22}$$

(6 marks)

Question 1c

$$\lim_{x \rightarrow 0} \left(\frac{\sin(3x) + \sin(4x)}{7x} \right)^{\frac{12}{x^2}}$$

(7 marks)

Question 1d

$$\lim_{x \rightarrow 0} x^5 \cos\left(\frac{1}{x^{26}}\right)$$

(7 marks)

Question 2a

Apply derivative tests to find the local and absolute maximum and absolute minimum of the piecewise function

$$f(x) = \begin{cases} x^2 - 2x + 3, & 0 \leq x \leq 2, \\ \frac{17}{2} - x - \frac{7}{x}, & 2 < x \leq 4, \end{cases}$$

on the interval $[0, 4]$.

(10 marks)

Question 2b

Define a function $H : (0, \infty) \rightarrow \mathbb{R}$ by $H(x) = \sqrt{x} + \ln x$. Use an appropriate Taylor polynomial for H at 26 to estimate $H(26.1)$ correct to 3 decimal places.

(10 marks)

Question 3

Solve and simplify the following:

$$\frac{d}{dx} \int_{x^3}^{e^{2x}} \ln(26 + t) dt.$$

(15 marks)

Question 4

Employ various techniques of integration to evaluate the following integrals.

Question 4a

$$\int \frac{3x + 1}{(x + 1)(x^2 - 2x + 26)} dx$$

(7 marks)

Question 4b

$$\int \frac{dx}{(26 - x^2)^{3/2}}.$$

(6 marks)

Question 4c

$$\int_0^{\sqrt{26}} \tan^{-1}\left(\frac{x}{\sqrt{26}}\right) dx$$

(7 marks)

Question 5a

Find the exact area of the region in the first quadrant enclosed by the curves

$$y = \frac{12}{x}, \quad y = x, \quad y = 3.$$

(10 marks)

Question 5b

Calculate the volume of the solid obtained by revolving the region in the first quadrant between the curves

$$y = \sqrt{9x} \quad \text{and} \quad y = 3x$$

about the y -axis. Give your answer in exact form.

(10 marks)

----- END OF PAPER -----